

Enterprise Telecom Fiber Churn Intelligence

Technical Design Document

Version 1.1.0

Date: June 17, 2026

Classification: Internal — POC

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1. Executive Summary

The Enterprise Telecom Fiber Churn Intelligence system is a proof-of-concept (POC) platform designed to predict customer churn for fiber broadband subscribers. Built on a three-tier microservices architecture, the system comprises a FastAPI-based ML inference service, a Spring Boot integration layer, and a React-based interactive dashboard served via Express.

The platform ingests synthetic customer data (8,000 records), trains a Random Forest classifier achieving an F1 score of 0.889 and combined score of 0.800, and exposes churn predictions through a unified REST API. A configurable threshold-tuning mechanism enables three campaign profiles — Retention, Balanced, and Conservative — allowing business teams to align churn interventions with operational cost tolerance.

2. System Overview

2.1 Purpose

Provide telecom operations teams with a real-time churn risk scoring engine that ingests customer attributes and returns a churn probability, risk level (HIGH/MEDIUM/LOW), and campaign-appropriate threshold decision. The system is designed for extensibility to production-scale workloads.

2.2 Key Capabilities

- Synthetic customer data generation (10,000 records) with realistic churn drivers
- Machine learning model training pipeline with automated hyperparameter tuning
- Threshold tuning for three business campaign profiles
- RESTful API for single-customer churn prediction
- Feature importance analysis for model interpretability
- Interactive dashboard with dataset overview, model performance, and live prediction feed

2.3 Technology Stack

Layer	Technology	Version / Config
ML Runtime	Python 3.14	scikit-learn, pandas, numpy, joblib
ML Inference API	FastAPI	v1.1.0, uvicorn
Integration Layer	Spring Boot	Java microservice on port 3333
Frontend UI	React 18 + Vite	Express server on port 4000
Data Format	CSV	fiber_customers_realistic.csv, 8,000 rows
Model Persistence	Joblib	churn_model.pkl, model_columns.pkl
Configuration	JSON	threshold_tuning.json, training_summary.json

3. Architecture & Component Diagram

The system follows a three-tier architecture with clear separation of concerns:

3.1 Tier 1: Data & Model Layer (Python)

The data generation script (`generate_dataset.py`) produces a synthetic CSV dataset with 15 customer attributes plus a churn label. The training pipeline (`train_model.py`) performs feature engineering, trains Random Forest and Extra Trees classifiers via randomized search with 3-fold stratified cross-validation, selects the best model by combined F1/ROC-AUC score, and serialises the model and feature columns to pickle files.

3.2 Tier 2: API & Integration Layer

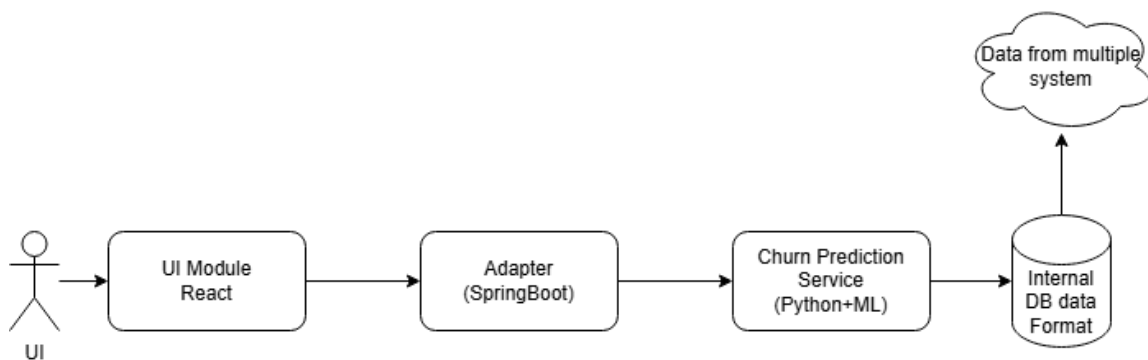
Two services comprise this tier:

- FastAPI : Serves the trained model via `/predict`, `/feature-importance`, and `/threshold-tuning` endpoints.
- Spring Boot: Acts as an enterprise integration layer, proxying predictions, providing batch churn data via `/api/churn/getchurndata`, and translating customer JSON for the ML service.

3.3 Tier 3: Presentation Layer (UI)

An Express serves a React SPA built with Vite. The dashboard fetches customer data from the local CSV, model metrics from FastAPI, and live prediction feeds from Spring Boot. A tabbed interface organises Overview, Model Performance, and Insights & Predictions views.

3.4 Request Flow



4. Data Pipeline Design

4.1 Data Generation

Generate 8,000 synthetic customer records with realistic churn drivers. Feature distributions approximate real-world telecom patterns:

Attribute	Type	Distribution
Plan	Categorical	Basic (25%), Plus (25%), Premium (25%), Ultra (25%)
monthlyPrice	Continuous	699, 999, 1499, 2499 (tied to plan)
tenureMonths	Integer	Uniform 1–60
Outages	Integer	Random 0–8
complaints	Integer	Random 0–7
supportCalls	Integer	Random 0–8
latePayments	Integer	Random 0–5
competitorAvailable	Binary	50% each
monthlyContract	Binary	50% each
speedMbps	Integer	100/200/500/1000 (tied to plan)
avgMonthlyUsageGb	Integer	N(180, 70), clipped ≥ 20
recentPlanChange	Binary	18% true
contractRenewalDue	Binary	25% true
Region	Categorical	North, South, East, West (uniform)
Churn	Binary	~30% positive rate via logistic churn scoring

4.2 Churn Scoring Formula

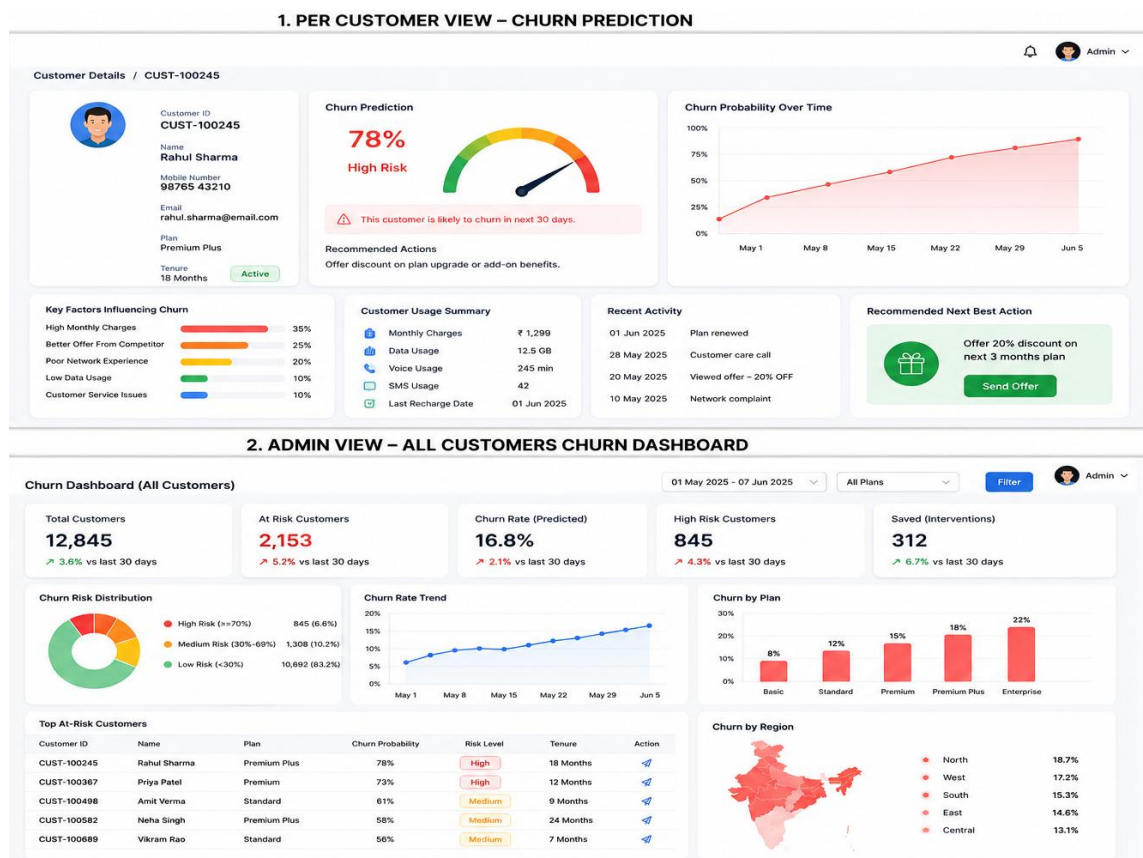
Each record receives a churn probability derived from a logistic function of weighted feature contributions:

$$\text{churn_score} = 0.12 \times \text{outages} + 0.10 \times \text{complaints} + 0.08 \times \text{supportCalls} + 0.14 \times \text{latePayments} + 0.15 \times \text{competitor} + 0.10 \times (\text{tenure} < 6) + 0.08 \times (\text{price} > 1200) + 0.12 \times \text{recentPlanChange} + 0.10 \times \text{contractRenewalDue} + 0.10 \times (\text{usage} < 80) + 0.03 \times (\text{monthlyContract and tenure} < 12) + N(0, 0.15)$$
$$\text{churn_probability} = 1 / (1 + e^{(-\text{churn_score})})$$

4.3 Training Pipeline

- Feature engineering: service_stress (outages+complaints+supportCalls), billing_risk (latePayments×price/100), price_per_mbps (price/speed), usage_drop (usage<80), tenure_bucket (new/short/mid/long)
- One-hot encoding of plan, region, and tenure_bucket → 28 features total
- Train/test split: 80/20 stratified by churn label
- RandomizedSearchCV with 2 iterations, 3-fold stratified CV, F1 scoring
- Model candidates: RandomForestClassifier, ExtraTreesClassifier
- Selection metric: $0.7 \times F1 + 0.3 \times \text{ROC-AUC}$ (combined score)
- Winner: RandomForest (n_estimators=250, max_depth=12, min_samples_leaf=2, class_weight=balanced_subsample)
- Threshold tuning across 17 thresholds from 0.10 to 0.90

5. Dashboard View



6. Machine Learning Model Design

6.1 Feature Importance (Top 8)

Feature	Importance Score
monthlyPrice	High
tenureMonths	High
latePayments	High
Outages	Medium-High
Complaints	Medium-High
service_stress	Medium-High
competitorAvailable	Medium
monthlyContract	Medium

7. Appendices

A. Data Dictionary

Complete attribute reference for the customer dataset and prediction request schema.

B. Glossary

Term	Definition
Churn	Customer cancellation of fiber subscription
Retention Campaign	Proactive outreach to at-risk customers before they churn
Precision	$TP / (TP + FP)$ — accuracy of positive predictions
Recall	$TP / (TP + FN)$ — ability to find all positive instances
F1 Score	Harmonic mean of precision and recall
ROC-AUC	Area under the receiver operating characteristic curve
Random Forest	Ensemble of decision trees trained on bootstrapped samples
StratifiedKFold	Cross-validation that preserves class proportions in each fold